THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 14: **Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The total surface area of a cuboid of length 10 cm, width 6 cm and height 4 cm is: 1

(a) 248 cm^2 **(b)** 240 cm^2 **(c)** 124 cm^2 **(d)** 134 cm^2

2. The volume of a cuboid measuring 12 cm $\times$ 10 cm $\times$ 5 cm is: 1

(a) 600 cm^3 **(b)** 620 cm^3 **(c)** 460 cm^3 **(d)** 27 cm^3

3. The volume of a cube is 125 cm^3. Its total surface area is: 1

(a) 150 cm^2 **(b)** 100 cm^2 **(c)** 75 cm^2 **(d)** 25 cm^2

4. Taking $\pi = \frac{22}{7}$, the volume of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 1540 cm^3 **(b)** 3080 cm^3 **(c)** 770 cm^3 **(d)** 440 cm^3

5. Taking $\pi = \frac{22}{7}$, the curved surface area of a cone of base radius 14 cm and slant height 15 cm is: 1

(a) 330 cm^2 **(b)** 1320 cm^2 **(c)** 814 cm^2 **(d)** 660 cm^2

6. A cone has base radius 5 cm and height 12 cm. Its slant height is: 1

(a) 12 cm **(b)** 14 cm **(c)** 13 cm **(d)** 17 cm

7. Taking $\pi = \frac{22}{7}$, the volume of a sphere of radius 10.5 cm is: 1

(a) 2425.5 cm^3 **(b)** 4851 cm^3 **(c)** 9702 cm^3 **(d)** 3638.25 cm^3

8. A pyramid has a square base of side 6 cm and a height of 10 cm. Its volume is: 1

(a) 120 cm^3 **(b)** 60 cm^3 **(c)** 360 cm^3 **(d)** 240 cm^3

9. If the radius of a sphere is doubled, its volume becomes: 1

(a) 4 times **(b)** 6 times **(c)** 2 times **(d)** 8 times

10. **Assertion (A):** If the radius of a sphere is doubled, its surface area is also doubled. 1
Reason (R): The surface area of a sphere is $4\pi r^2$.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the volume of a hemisphere of radius 10.5 cm ($\pi = \frac{22}{7}$). 2

12. Write the formulas for the surface area and the volume of a sphere of radius r. 2

13. Why does the formula for the volume of a cone contain the factor $\frac{1}{3}$? How can this be shown by an experiment? 2

14. How many cubes of side 2 cm can be cut from a cuboid of 10 cm $\times$ 8 cm $\times$ 6 cm? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A metal sphere of radius 3 cm is melted and recast into small cones of radius 1 cm and height 3 cm. How many cones are made? 3

16. A conical tent has base radius 7 m and height 24 m. Find the canvas needed and its cost at ₹120 per m^2 ($\pi = \frac{22}{7}$). 3

17. The curved surface area of a cone is 462 cm^2 and its slant height is 21 cm. Find the base radius and the total surface area ($\pi = \frac{22}{7}$). 3

OR

A right-angled triangle with sides 7 cm, 24 cm and 25 cm is rotated through 360° about the side of 24 cm. Find the volume and the curved surface area of the solid formed ($\pi = \frac{22}{7}$). 3

18. A conical cup of height 18 cm is filled with water up to a depth of 6 cm (measured from the apex). What fraction of the cup's volume is occupied by water? 3

OR

A pyramid has a square base of side 9 cm and height 10 cm. Find its volume. How does it compare with a prism of the same base and height? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Show how Archimedes' cylinder gives the surface area of a sphere.
(ii) For a sphere of radius 10.5 cm find the surface area and the volume ($\pi = \frac{22}{7}$). 5

OR

Case study: Three identical metal spheres, each of radius 3.5 cm, fit exactly in a row inside a closed cuboidal box. 5
(i) Write the internal dimensions of the box.
(ii) Find the volume of the box.
(iii) Find the total volume of the three spheres.
(iv) The remaining space is filled with preservative liquid; find its volume ($\pi = \frac{22}{7}$).

20. A solid toy is a cone of base radius 10.5 cm and height 14 cm with a hemisphere of the same radius fixed on its base. ($\pi = \frac{22}{7}$) 5
(i) Find the slant height of the cone.
(ii) Find the total outer surface area of the toy.
(iii) Find the total volume of the toy.

OR

Case study: A golgappa vendor's cylindrical vessel has radius 12 cm and holds spiced water to a height of 35 cm. Assume each golgappa is a sphere of radius 2.1 cm and takes pani equal to 50% of its volume ($\pi = \frac{22}{7}$). 5
(i) Find the volume of pani in the vessel.
(ii) Find the volume of one golgappa.
(iii) Find the pani used in one golgappa.
(iv) Estimate how many golgappas can be served.
(v) Why would a different assumption change the answer?

END OF QUESTION PAPER — **Total: 40 Marks**